Time series: Introduction

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Lecture notes 1

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Homework is given on the last slide

Time series: overview

Data gathered sequentially in time are called a *time series*.

What can we do with a time series?

- Smoothing
- Decomposition
- Change-point detection
- Model fitting

Identifying a suitable model (AR, MA, ARMA, ARIMA, etc.) that describes a given time series and estimating its parameters.

- Forecasting

Continuation/prediction of the time series using the estimated model.

For the model "observed series=signal+noise", we usually want to estimate the signal and forecast it.

Syllabus

- Various models of time series
- Autocovariance, autocorrelation, spectral density
- Prediction equation in time domain
- Prediction bounds for normal processes
- Autoregressive (AR), moving average (MA), ARMA models
- Sample theory
- Yule-Walker equations
- Model identification and estimation in time domain
- ARIMA models
- Singular Spectrum Analysis (SSA)

Brockwell P.J., Richard A.D. Introduction to time series and forecasting. Taylor & Francis, 2002. (*available as pdf-file*)

Learning Outcomes

- Fit models for a large variety of data from economics and social sciences.
- Apply modern methods of statistical inference.
- Analyze and forecast using ARMA and ARIMA models.
- Apply the R package for time series analysis.
- Utilize AI tools to assist in learning R programming.

Assessment, timetable, programming

Examination: 100%

Timetable
Mon 17:10-18:00, room Abacws/5.05, weeks 2,4,6,...
Tue 11:10-13:00, room Abacws/5.05

The R package can be downloaded from

`http://www.r-project.org/`
`http://sourceforge.net/projects/rportable/`

The most convenient tool is **Rstudio**
`https://www.rstudio.com/`

Use AI for R programming for time series

AI tools enable you to learn R programming and time series analysis.

Compare responses from at least two AI tools

- Copilot
- Gemini
- Claude
- Grok
- Deepseek
- Kimi K2

AI is good at responding to 'simple' questions and can be wrong in responding to 'difficult' questions. Specify a level of response (e.g. student, expert, lyrics writer).

Resources on how to use AI

How to use the AI-powered chatbot to learn anything you want

`https://www.studysmarter.co.uk/magazine/chatgpt-for-studying/`

Studying 6 ways to use ChatGPT for studying

`https://transcribe.com/blog/how-to-use-chatgpt-for-studying`

A Student's Guide to Writing with ChatGPT

`https://openai.com/chatgpt/use-cases/student-writing-guide/`

215+ ChatGPT Prompts + How to Write your Own

`https://writesonic.com/blog/chatgpt-prompts`

Top 120 ChatGPT Prompts for Math Related Problems

`https://remagineai.com/chatgpt-prompts/math-chatgpt-prompts-guide`

Ask AI on how to use AI for a particular problem.

Areas of science with time series

- Economics and Finance
- Environmental and Pollution Modelling
- Meteorology and Hydrology
- Demographics
- Medicine
- Engineering
- Quality Control
- Sports

Time series are typically observed at equally spaced intervals in discrete time: at each second, hour, day, month, year, etc.

Detailed examples 1

- In **business**, we observe daily stock prices, weekly interest rates, quarterly sales, monthly supply figures, annual earnings, etc.
- In **agriculture**, we observe annual yields, crop production, daily crop prices, annual herd sizes, etc.
- In **engineering**, we observe electric signals, voltage measurements, etc.
- In **natural sciences**, we observe chemical yields, turbulence in ocean waves, Earth's tectonic plate movements, etc.
- In **sports**, we observe betting odds movements, etc.

Detailed examples 2

- In **medicine**, we observe EKG measurements on patients, drug concentrations, blood pressure readings, etc.
- In **epidemiology**, we observe the number of flu cases per day, the number of health-care clinic visits per week, annual tuberculosis counts, etc.
- In **meteorology**, we observe daily high temperatures, annual rainfall, hourly wind speeds, earthquake frequency, etc.
- In **social sciences**, we observe annual birth and death rates, accident frequencies, crime rates, school enrollments, etc.

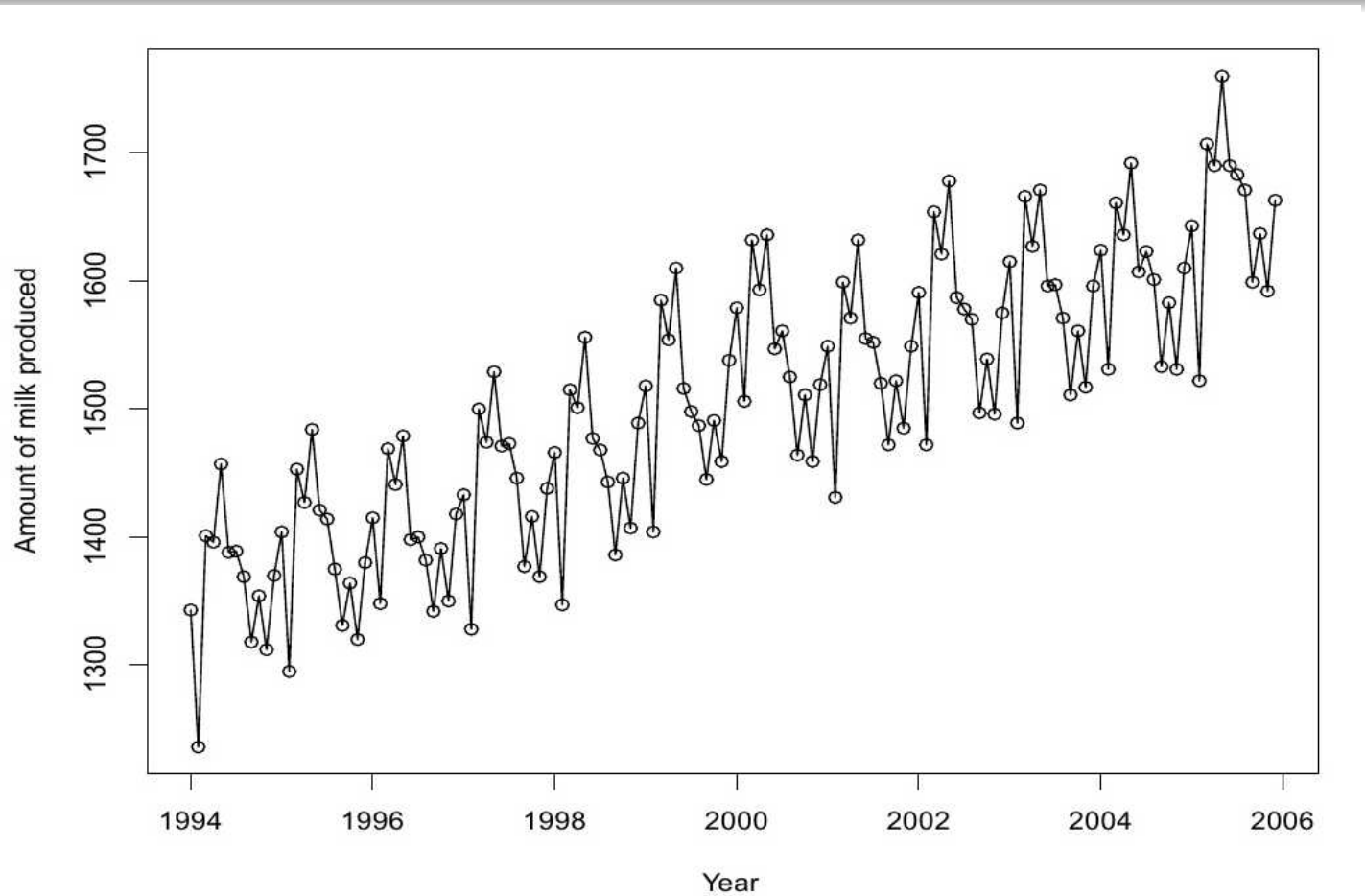
Objectives of time series analysis

- **Interpretation**, e.g. seasonal adjustment.
- **Forecasting**, e.g. predict unemployment.
- **Control**, e.g. impact of monetary policy on unemployment.
- **Hypothesis testing**, e.g. global warming.
- **Simulation**, e.g. estimating probability of catastrophic events.

Importance of forecasting

- **Governments** forecast unemployment rates, interest rates, and expected revenues from income taxes for policy purposes.
- **Marketing executives** forecast demand, sales, and consumer preferences for strategic planning.
- **College administrators** forecast enrollments to plan for facilities and for faculty recruitment.
- **Retail stores** forecast demand to control inventory levels, hire employees and provide training.
- **Insurance managers** forecast portfolio risk.

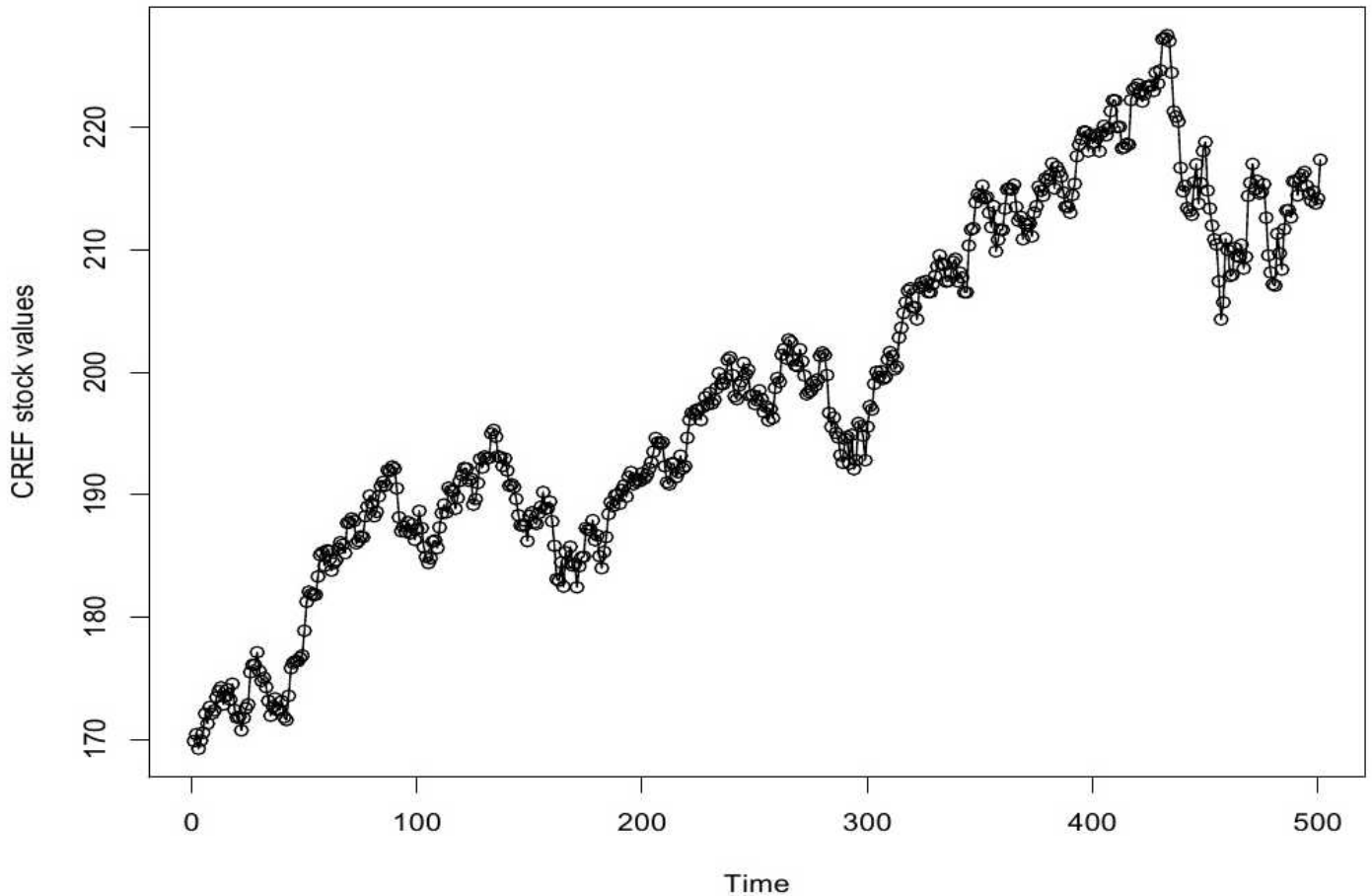
Data 1: milk production



United States milk production data, monthly, measured in millions of pounds, from January, 1994 to December, 2005, $n = 144$.

What are the noticeable patterns?

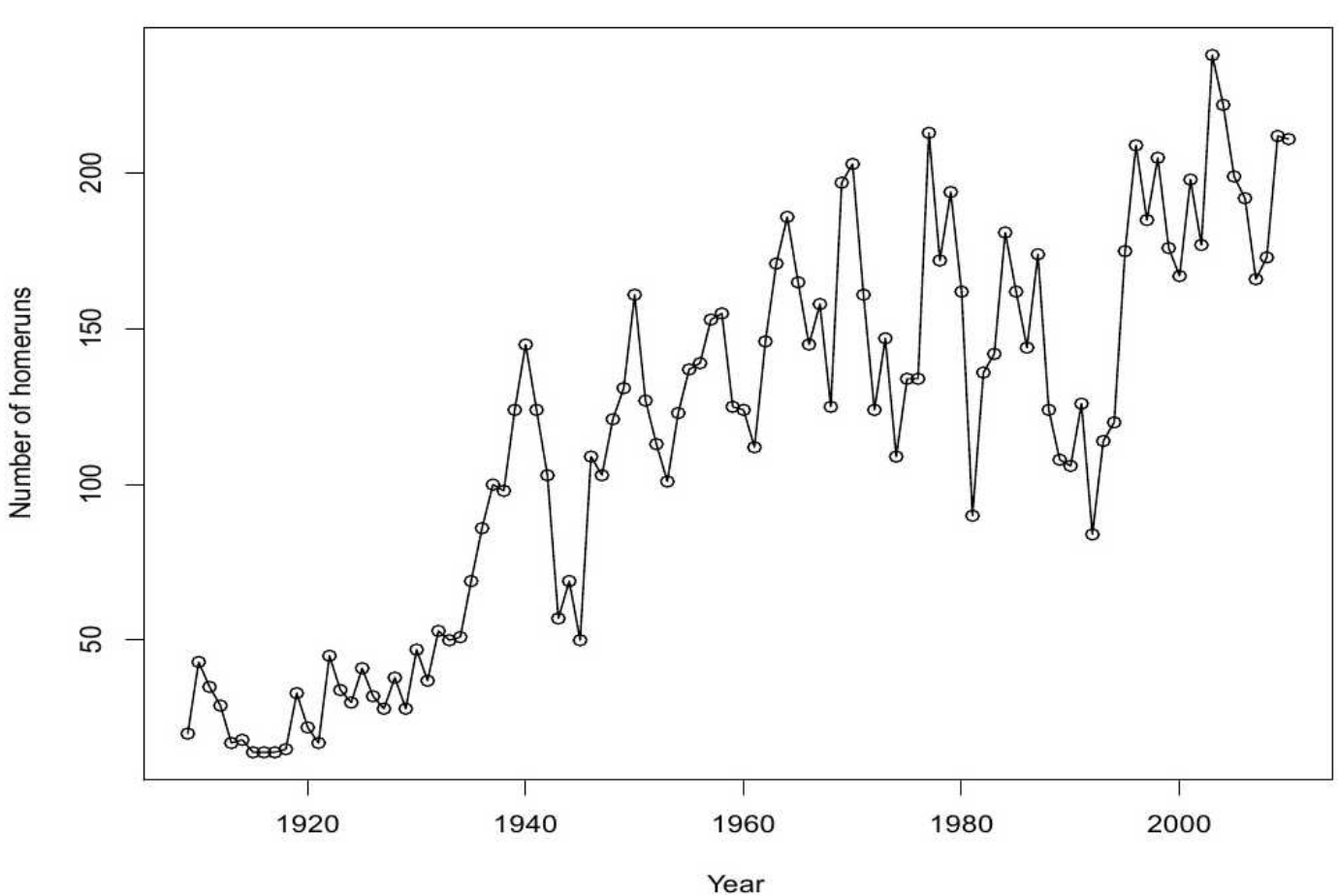
Data 2: stock data



CREF stock data. Daily values of one unit of CREF stock values: August 26, 2004 to August 15, 2006, $n = 501$.

What are the noticeable patterns?

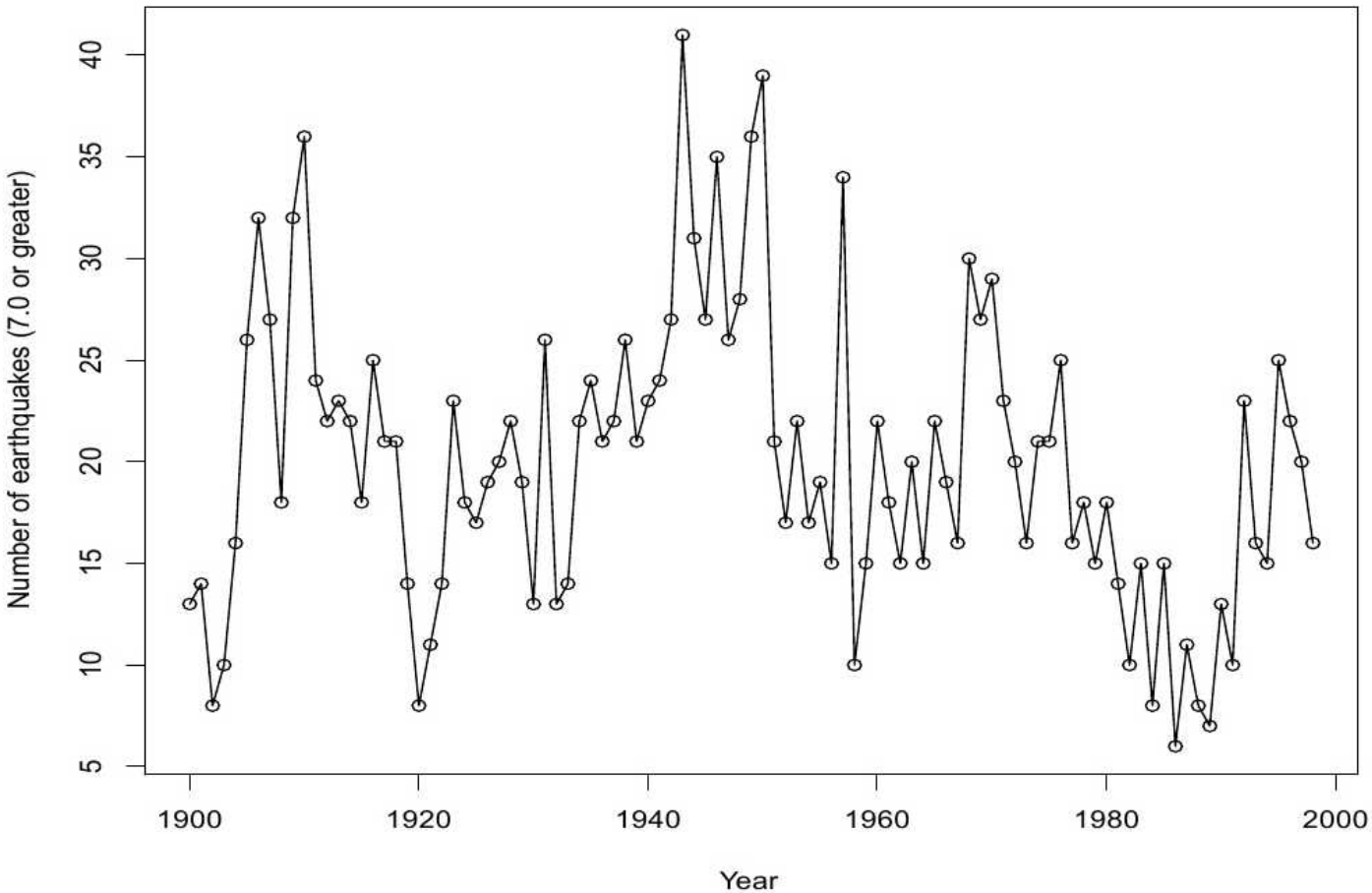
Data 3: homerun data



Homerun data, annually. Number of homeruns hit by the Boston Red Sox, a professional baseball team, each year during 1909-2010, $n = 102$.

What are the noticeable patterns?

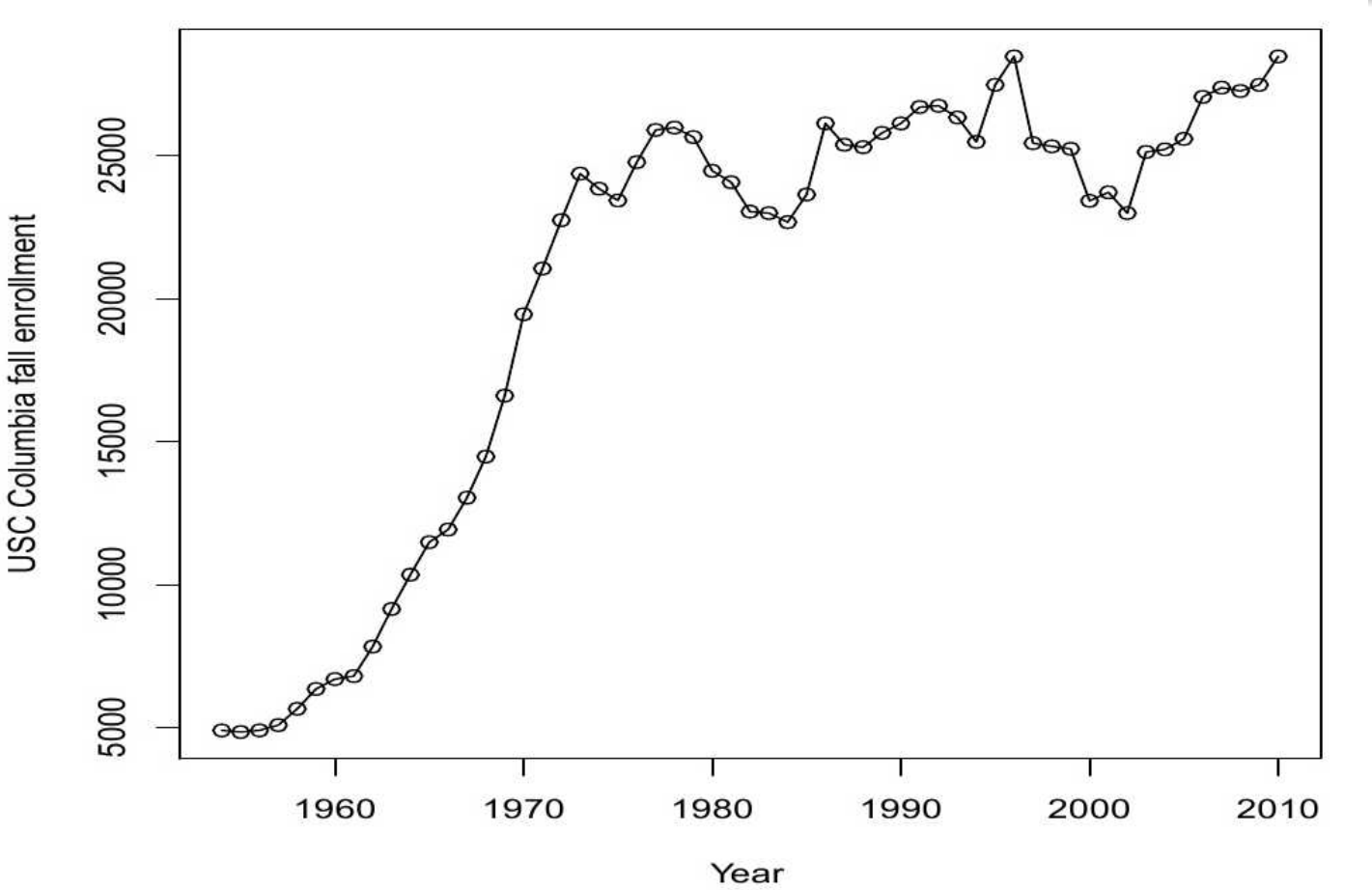
Data 4: earthquake data



Earthquake data, annually. Number of 'large' earthquakes per year from 1900-1998, $n = 99$.

What are the noticeable patterns?

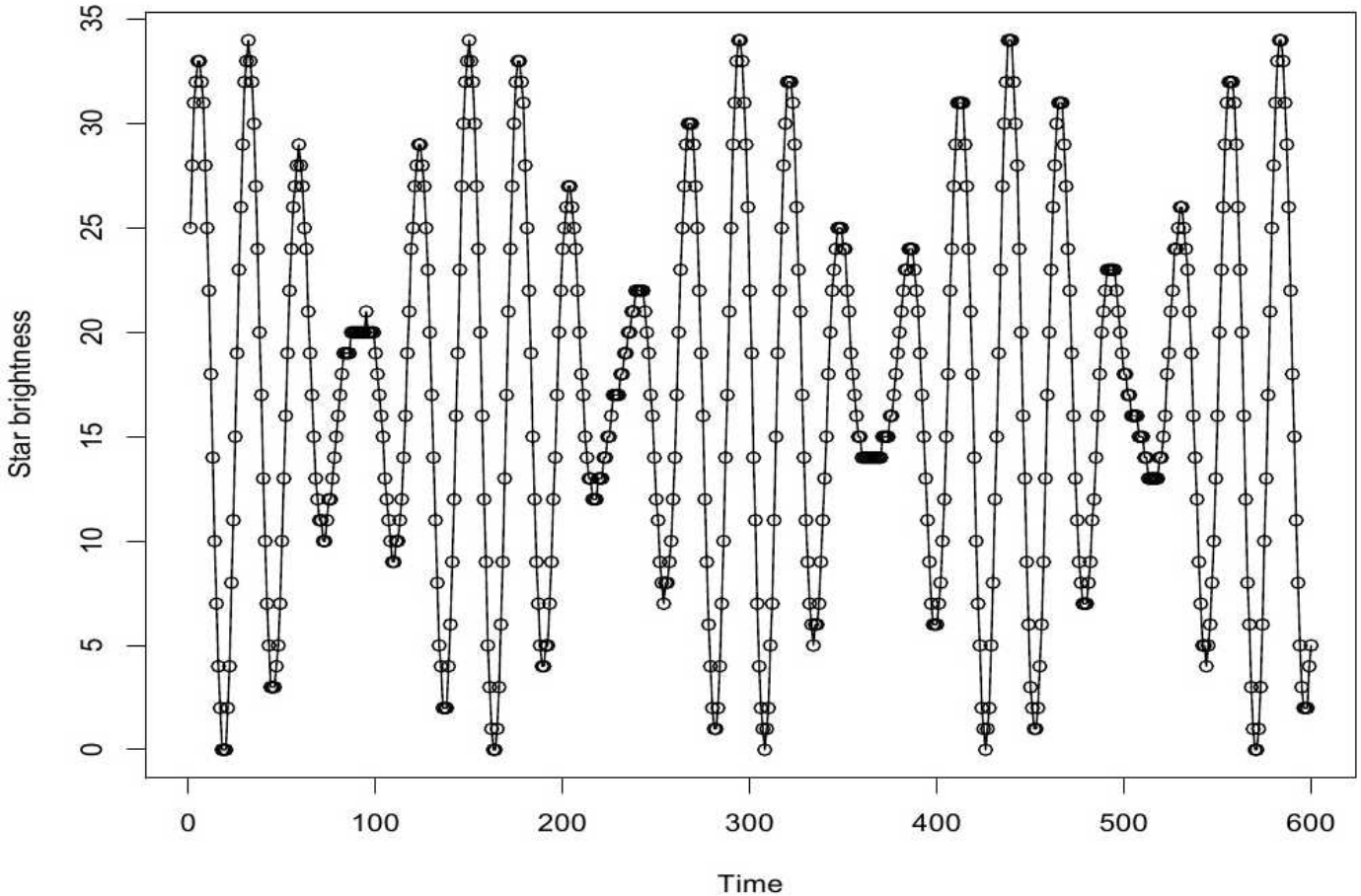
Data 5: enrollment data



University of South Carolina fall enrollment data, annually. Number of students registered for classes on the Columbia campus during 1954-2010, $n = 57$.

What are the noticeable patterns?

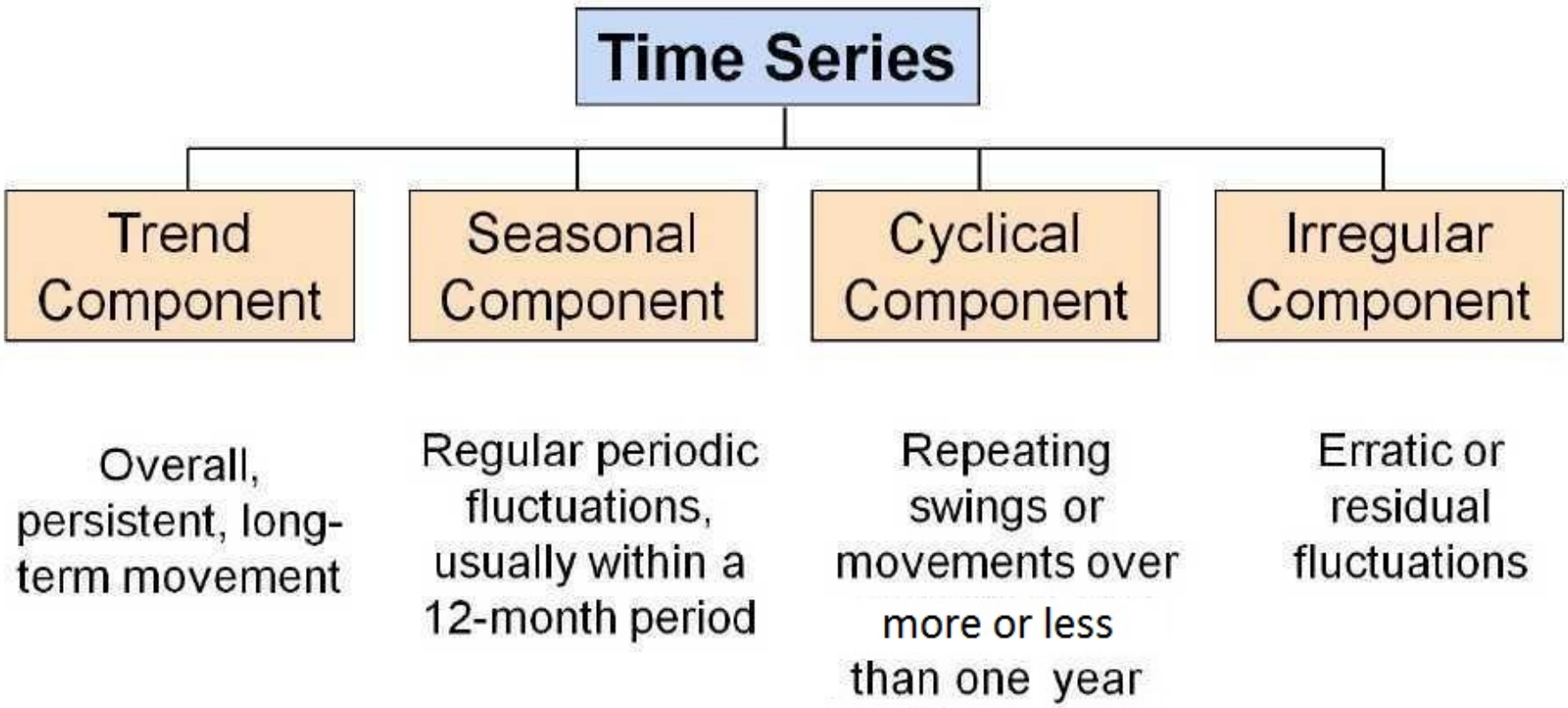
Data 6: star brightness



Star brightness data, daily. Measurements for a single star taken over 600 consecutive nights, $n = 600$.

What are the noticeable patterns?

Components of time series



Homework

- Explore AI tools and gain a general overview of time series analysis.
- Watch introductory videos on the R package:
  `https://www.youtube.com/watch?v=LjuXiBjxryQ`
  `https://www.youtube.com/watch?v=7cGwYMhPDUY`
- Download and install the R package and Rstudio.
- Perform some arithmetic calculations, i.e. use R as a calculator.
- Create several variables and perform operations with them.
- Create vectors and perform operations with them.

® *v1=1+4; print(v1); v2=exp(2)+1; print(v2)*
a=2; b=5+log(2); z=a+b; print(z)
w1=c(1,7,3); w2=c(3,1,4); q=w1+w2-1;
*ind=c(1:20); y=sin(ind*0.3); plot(ind,y);*
sumind=sum(ind); maxy=max(y);
print(sumind); print(maxy); print(y[y>0.4]);